CampusPay X

Self-Healing Software — Resilience Report

Test environment: Node.js/Express, MongoDB Atlas, PM2, and a simulated payment gateway. **Test date:** 26 October 2026. **Scope:** local development demonstration; no real payments.

1. Objective and system design

CampusPay X models a campus canteen payment workflow where an ambiguous payment outcome must not be treated as a failure or success without evidence. The system persists transactions and mock-gateway outcomes, validates responses, classifies failures, and reconciles eligible pending transactions. PM2 supervises the Node.js process, while application recovery is responsible for restoring transaction state after restart.

Architecture

Browser Control Center
Payment API (Express) → Input validation / idempotency
Retry policy + circuit breaker → Simulated gateway
MongoDB Atlas: transactions + authoritative mock outcomes
Reconciliation worker / startup scan → incidents + reliability events
PM2 process supervisor restarts Node.js after unexpected exit

Recovery safety principle

A gateway response is not trusted solely because it exists. Reconciliation checks the persisted authoritative simulated outcome and relevant transaction details. If no authoritative outcome can be verified, the safe state is pending—not fabricated success. PM2 restart and payment recovery are separate operations.

The environment is simulation-only and does not connect to a real bank or payment provider.

2. Failure modes, detection and healing

The table distinguishes observed screenshot evidence from mechanisms documented in the code. Only the lost-response benchmark has a measured per-transaction recovery duration in the supplied output. Other recovery times are not invented.

Failure	Detection	Recovery mechanism	Measured recovery time / evidence
Slow gateway	Request timeline shows gatewayMode SLOW; request remains in the resilience path until response arrives.	Controlled delay expires; response is validated and transaction is finalized.	Screenshot shows request at 23:13:19 and verification at 23:13:24 (about 5 seconds wall-clock at displayed timestamp precision). Exact duration not shown.
Lost response	Outcome uncertain; FAILURE_CLASSIFIED: GATEWAY_RESPONSE_LOST and reconciliation-required events appear.	Look up persisted mock-gateway outcome; reconcile the original transaction; emit TRANSACTION_RECONCILED and RECOVERY_VERIFIED.	1,342 ms recoveryDurationMs in the supplied event screenshot. Benchmark sample: 228, 228, 227 ms (mean ≈227.7 ms). These are different measurements.
Invalid response	INVALID_GATEWAY_RESPONSE and RESPONSE_VALIDATION_FAILED events; state shown as PENDING while outcome is checked.	Reject malformed response; use authoritative persisted outcome for reconciliation when available.	Screenshot shows event sequence around 23:14:30; no exact recoveryDurationMs is visible. Exact duration not established.
Gateway outage	Gateway unavailable events, retry attempts and circuit-breaker OPEN state / cooldown.	Bounded retries, automatic mock-fault expiry, then half-open probe; close breaker only after successful probe.	No complete timestamped outage-to-CLOSED evidence supplied in this set. Not reported as a measured recovery time.
Backend crash	PM2 detects Node process exit; startup logs and health endpoint indicate process recovery.	PM2 restarts Node; startup recovery reconciles eligible pending transactions and verifies the original transaction.	Crash-test output with start/end timestamps was not included in the supplied evidence. Not reported as a measured recovery time.

3. Before-and-after benchmark

The supplied benchmark run was created at 2026-10-26T17:49:31.744Z (26 October 2026, 23:19:31.744 IST). Run ID: 2b92680c-53db-495d-8d1e-810580f2a9a3. Scenario: LOST_RESPONSE. Sample size: 3.

Metric	Baseline at measurement	Self-healing cohort
Transactions attempted	3	3
Recovered at measurement	0/3 (0%)	3/3 (100%)
Pending at measurement	3	0
Cohort elapsed time	778 ms	1,603 ms
Per-transaction recovery durations	No recovery by measurement point	228, 228, 227 ms

Observed change

At the benchmark measurement point, immediate reconciliation changed the recovery rate from 0% to 100% (+100 percentage points) and the pending count from 3 to 0. The mean of the three recorded self-healing recovery durations is (228 + 228 + 227) / 3 = 227.7 ms, approximately.

Important qualification

The baseline cohort is measured before immediate reconciliation, while the self-healing cohort is reconciled immediately against persisted simulated gateway outcomes. The benchmark itself states that periodic recovery may later heal baseline transactions. Therefore this is a point-in-time experiment, not a claim that baseline transactions are permanently unrecoverable or that every production recovery will take 227.7 ms.

Additional cumulative reliability summary

At the time of the supplied summary, the database contained 207 transactions, 204 successful, 3 pending, 157 incidents, 154 recovered incidents, and 3 unresolved incidents. The summary reported a 98.1% recovery success rate across 154 measured recovery samples. Its average duration was 24,813,840 ms and p95 was 111,737,256 ms. Because this is a cumulative dataset

containing older test records and long-lived pending records, these aggregate durations are not used as the clean before/after benchmark.

No unmeasured outage or crash recovery time is inferred from the benchmark.

4. Known limitations and submission evidence

Known limitations

- **Mock gateway only:** this experiment does not establish resilience against a real payment provider, real banking systems, or arbitrary internet failures.
- **Unknown outcome:** when no authoritative gateway outcome is available, a transaction remains pending until safe retry or reconciliation is possible. The application must not invent success.
- **In-memory circuit breaker:** breaker state resets when the Node process restarts. A multi-instance production deployment needs explicit shared or per-instance semantics.
- **Separate process and data recovery:** PM2 can restore the Node process, but this alone does not prove that a pending payment was reconciled. Both health restoration and transaction-level recovery evidence are required.
- **Database dependency:** this test does not demonstrate recovery from a prolonged MongoDB Atlas outage or loss of authoritative gateway records.
- **Development-only controls:** chaos endpoints must remain disabled in production. Benchmark and chaos endpoints create simulated records and should only be used on a local development environment.

Honesty Clause — A Failure CampusPay X Cannot Fully Heal Yet

CampusPay X can detect gateway-related failures and use recovery mechanisms such as retries and circuit-breaker protection. However, if the simulated gateway remains unavailable beyond the configured retry and recovery limits, the system cannot restore the gateway itself. The circuit breaker can stop repeated failing requests, but it cannot make the unavailable service respond.

In this situation, a transaction may remain in a PENDING state until the gateway becomes available and its outcome can be safely reconciled. The system must not assume that a timeout means payment failure or mark an uncertain transaction as successful without verification.

Why it cannot fully heal this failure yet: The application controls its own retry and recovery logic, but it cannot guarantee the availability of an external dependency. If no reliable gateway response or reconciliation result is available, automatic recovery has reached its limit.

Planned improvement: Improve half-open circuit-breaker probing, retry scheduling, pending-transaction reconciliation, and incident escalation so the system can resume safely when the gateway recovers and clearly report cases requiring further investigation.

We acknowledge that self-healing has limits. Our verified PM2 crash-recovery test demonstrates recovery from a process crash, but it does not prove that every gateway failure can be healed automatically. We report this limitation transparently rather than claiming complete recovery in every situation.

Evidence included in the supplied materials

Evidence	What it supports	Status
Normal payment screenshots	SIMULATED_SUCCESS, transaction ID, persistence in UI/Atlas	Provided
Slow gateway screenshots	SLOW mode and successful verification; displayed times about 5 seconds apart	Provided
Lost-response timeline	Failure classification, reconciliation and RECOVERY_VERIFIED; 1,342 ms recoveryDurationMs	Provided

Invalid-response timeline	Invalid response and validation failure; pending/reconciliation sequence	Provided
Gateway outage recovery	Breaker OPEN → half-open probe → CLOSED and verified transaction	Provided
PM2 crash recovery	PM2 restart, healthy endpoint, original transaction success and RECOVERY_VERIFIED	Provided

PM2 crash recovery





